



SECP2523: DATABASE

SEM I 2024/2025

DATABASE DESIGN DESCRIPTIONS: CONCEPTUAL DATABASE DESIGN FOR

KADA's Cooperative Web-Based System

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CHAPTER 1: Introduction

1.1 Overview about the company

The Kemubu Agricultural Development Board (KADA) is an institution within the Ministry of Agriculture and Food Security that promotes and encourages the economic and social growth of farmers in the Kemubu region. Within KADA there is an organization called KADA's staff cooperative whose members unite voluntarily to achieve common interests in the economic, social and cultural fields. The cooperative only allows employees that work under KADA to join as a member of the cooperative. The cooperative offers benefits such as loan services for business purposes, emergencies and welfare funds for its members.

1.2 Problems

Inconsistency in Data Entry

KADA's data management issues are a significant concern, primarily due to inconsistent data formats and reliance on manual data entry. These inconsistencies can cause significant errors, as data entered manually is prone to human error. It is also inefficient to always fill out repetitive information such as employees' names and ID numbers for applications.

Data Management Issues

KADA's cooperative uses paperwork as the main medium for membership and loan applications. Large volumes of documents can be hard to handle as they can be misplaced or lost during transfer or storage. Storage for the application forms also raises a problem as physical storage of documents requires space and that can lead to organisational challenges. Documents are also hard to retrieve when they need to be reviewed. It also raises security and privacy issues as sensitive information is involved

Limitations in Data Collection and Analysis

The manual method of collecting data will cause the data not to be categorised and not organised well. KADA's cooperative expresses their difficulty in generating monthly and

yearly reports of profits and losses. The lack of data analysis due to insufficient data can affect the corporation's decision-making process for prospects.

Reliance on Human Resources

The manual process heavily relies on large amounts of human resources as the whole membership and loan application process is extensive, making it labour-intensive and contributing to more costs. It also makes it more prone to human errors as various entities handle the entire process. There could be errors when reviewing applications to make sure the information provided is valid and failure to record payment details.

1.3 Proposal

The proposed system will be a website as an alternative to the manual system currently used by KADA. The system will still retain some functionality but with additional modifications that increase efficiency to enhance the working experience of the entities involved. The main improvement is digitising most of the processes of the current system such as filling in the membership and loan application. However, some system processes are intended to be kept the same, such as the board of directors meeting that KADA desires to be done online. An Information Systems expert as the admin will oversee the operation and maintain the system if needed. The website will have a database that users of the system can utilize to create, read, update, and delete data in the database. This addresses the data management issues where important data is stored in a safe place with a database. Only authorised personnel can have access to the data in the database.

Firstly, for existing employees of KADA who desire to apply for a membership application to be part of the cooperation when filling out the form, the system will have a function that controls data incompatible issues. For example when filling out the section that strictly uses numbers, the empty fields require an input of integers. If the user happens to input a value besides integers, the system will alert the user for a valid input instead. This is to combat inconsistent data entry issues. All payment processes will also be done through the website. A statement will be sent to the applicant as proof of successful payment. The

applicants can also view their application status to see if they are accepted or rejected as cooperative members through the system.

As for the employees of KADA that are members of the cooperative, when applying for a loan, they only need to provide information not in the existing database. Once they have provided relevant information, they can proceed with the payment process. The website also displays a chart of types of loans and their respective calculations according to the type of loan and length of the loan application. Only members that have at least RM 300 in their share modal are entitled to apply for loans. A processing fee of RM50 is charged each time an employee applies for a loan. A statement and receipt will be generated that can be viewed by the user on the website. The applicants can also track their application status, the website will display the current process undergoing for loan approval. The system will provide the same feature to overcome the data management issue for the loan application.

From the viewpoint of the board of directors, they can access the applicant's information from the website. From there, they can approve the loan applications. Since the approval can be done within the website, this saves a lot of time because they can do it anywhere and anytime. Lastly, the admin is in charge of the cooperative's policies such as updating the current loan rates, minimum amount of loan allowed. The admin can also view all the details of members inside the system such as loan repayments and fees so that a financial statement can be generated for the members reference.

1.4 Definitions, Acronyms and Abbreviations

Acronyms and Abbreviation	Term	Definition
BOD	Board of Directors	The Board of Directors are responsible for the authorization of the loan and membership application approval.
ERD	Entity Relationship Diagram	A type of flowchart that illustrates how “entities” such as people, objects or concepts relate to each other within a system.
ID	Identification	The action or process of identifying someone or something or the fact of being identified.
KADA	Kemubu Agricultural Development Board	An institution within the Ministry of Agriculture and Food Security that promotes and encourages the economic and social growth of farmers in the Kemubu region.

CHAPTER 2: Database System Requirements

2.1 System Overview

This system is a web-based solution for KADA's cooperative that streamlines the application process in KADA. The process includes both member and loan application, approval of both applications, financial management of loan offered by KADA.

2.2 Module <Membership>

2.2.1 Module Description

In this module, it will involve the users for in this case are employees who are going to register themselves as a member of the KADA. The employee needs to enter all the required information in the application form. After completing the registration, the Board of Directors will review and approve or reject the application. Their current member status can be checked at the website.

2.2.2 Transaction Requirement: Data Entry

1. Enter user's informations

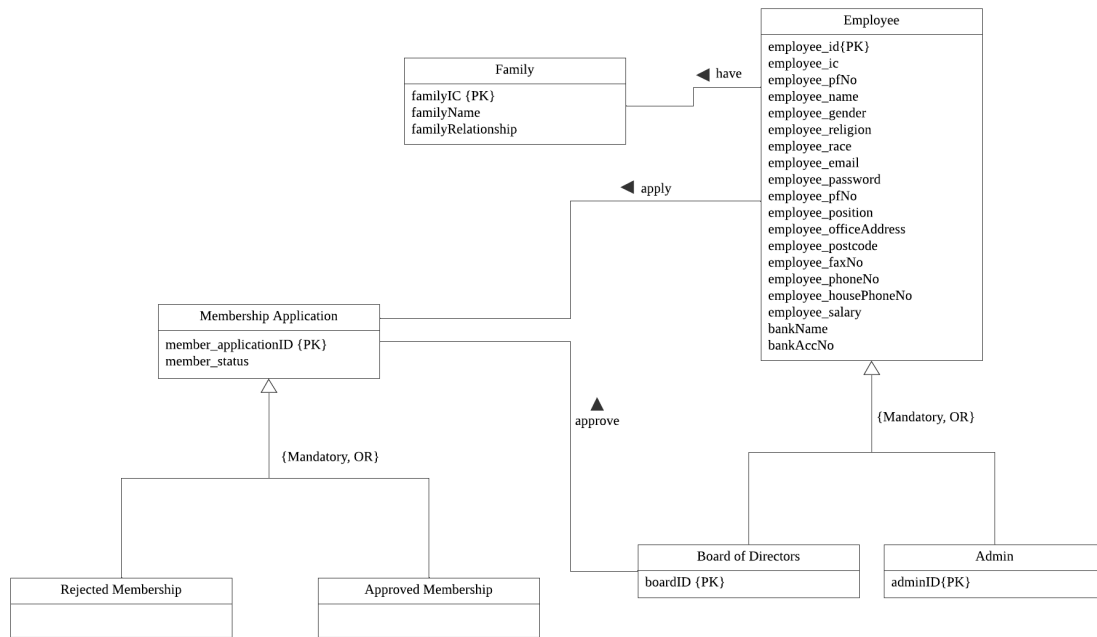
2.2.3 Transaction Requirement: Data Update/Deletion

1. Update user's information
2. Update current status of application

2.2.4 Transaction Requirement: Data Queries

1. List of the applications
2. List of user's information
3. List of user who got approve to be member
4. List of user who got rejected to be member

2.2.5 Local Conceptual Data Model (ERD) for Module 1



2.3 Module <Loan>

2.3.1 Module Description

In this module, the users involved are the employees of KADA that registered as the membership and board of directors. Before applying for a loan, the employee of KADA must be registered as a member. To apply for a loan, users need to give their information and loan details. After the users complete the loan application, the Board of directors will approve the loan application with updating the loan application status in the database. To approve the loan, the board of directors will need a list of information in users' databases.

2.3.2 Transaction Requirement: Data Entry

1. Enter loan details (loan amount etc)
2. Enter users' information

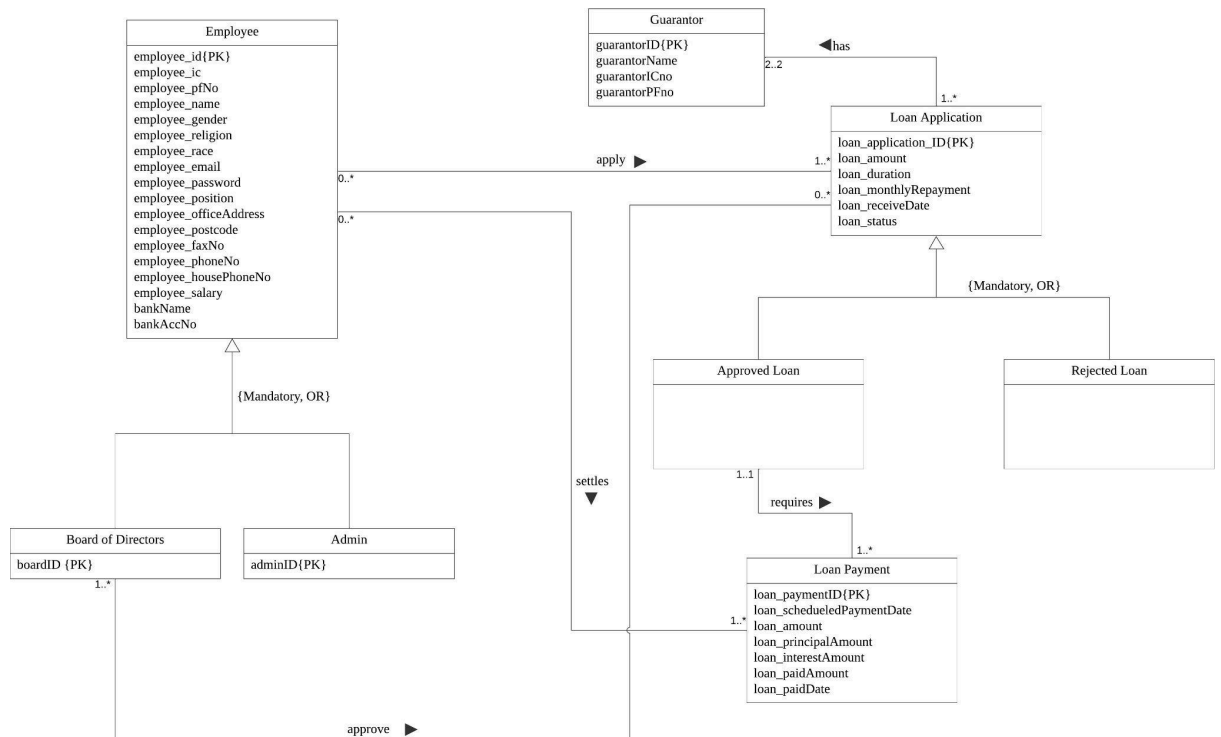
2.3.3 Transaction Requirement: Data Update/Deletion

1. Update users' loan application status
2. Update users' loan application

2.3.4 Transaction Requirement: Data Queries

1. List of users' information
2. List of loan application details
3. List of guarantors's information
4. Identify loan details

2.3.5 Local Conceptual Data Model (ERD) for Module 1



2.4 Module <Financial>

2.4.1 Module Description

This module involves the users of the system which are the employees and the members of the cooperative. Registering as a member of the cooperative requires the user to pay a registration fee. Employees can view their financial status if there are any outstanding balances after registering to become a member. They can update their financial status by providing a proof of payment that will be reviewed by the administrator of the system. Once the payment is acknowledged by the administrator, their financial status will be updated. They can also verify their financial status to acknowledge if all the details are all true.

For the members, they can also view their financial status to know whether or not they have any outstanding balance from monthly fees and any loans they applied for. They can view the progress of their loan repayment. They can also update their financial status by uploading proof of payment so that they can be reviewed by the administrator. Once the payment is acknowledged by the administrator, their financial status will be updated. They can also verify their financial status to acknowledge if all the details are all true such as their accumulated member shares.

For the administrators, they can review all the proof of payments submitted by the employees and members. They would have to verify if all the amounts paid are all correct. Once the payments have been verified, the administrator will approve the payments and the system will update the financial status of members and employees. The system generates a financial statement in real time for all members detailing their loan progress and member shares.

2.4.2 Transaction Requirement: Data Entry

1. Enter membership payment details
2. Enter loan payment details
3. Enter monthly fee payment details

2.4.3 Transaction Requirement: Data Update/Deletion

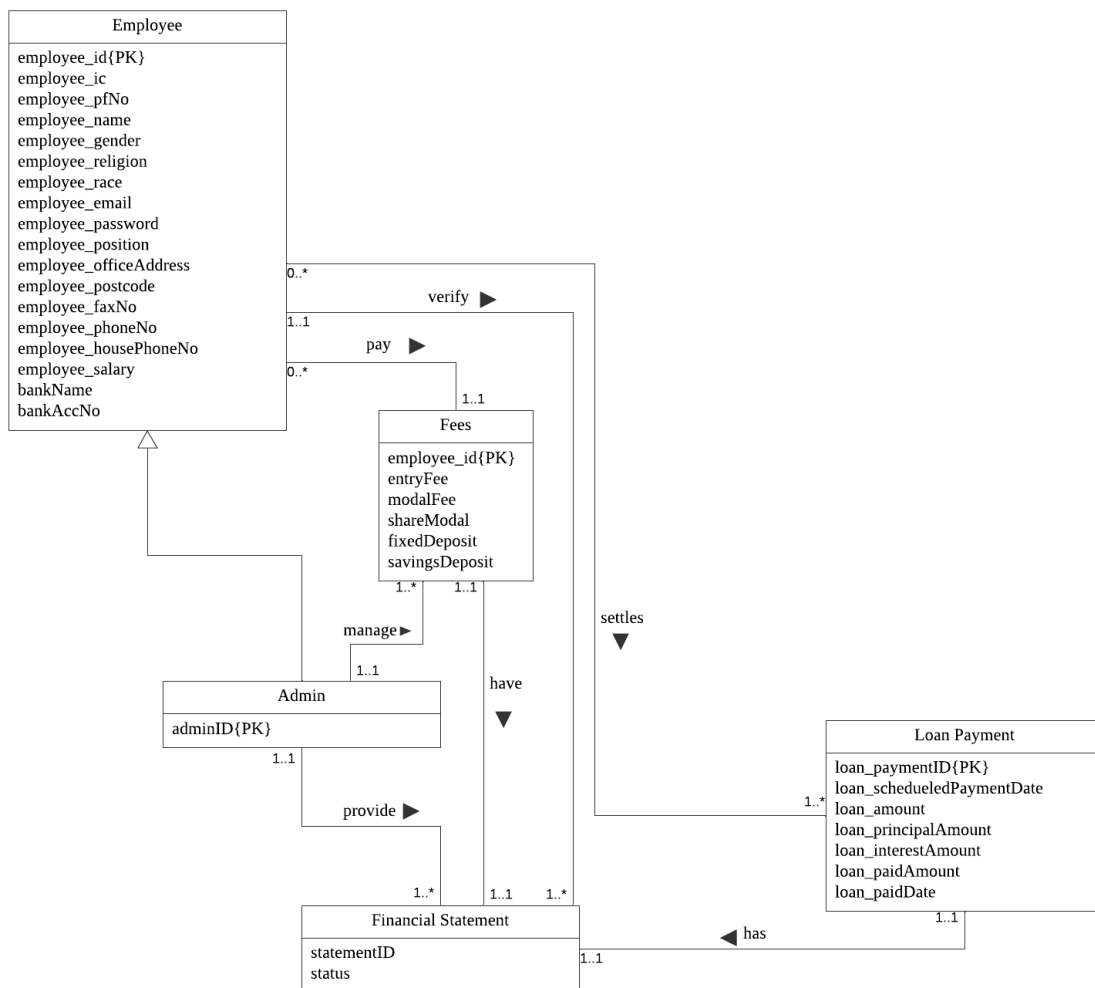
1. Update delete membership payment details
2. Update delete loan payment details
3. Update monthly fee payment details

2.4.4 Transaction Requirement: Data Queries

1. List the amount of loan that has not been paid
2. List the amount of loan that has been paid

3. Identify the total amount of fees based on monthly fee and loan amount per month if any.
4. Identify the total number of members that has not paid their monthly fees and loan amount per month if any
5. List the members that have uploaded their proof of payment
6. List the members that their payment have been acknowledged
7. Display the accumulated member's share amount

2.4.5 Local Conceptual Data Model (ERD) for Module 4



2.5 Module <Administration>

2.5.1 Module Description

The system's administrator will be able to log into the system so that the system's administrator can manage the cooperative's database and policies. The administrator can create new loan policies such as a new type of loan that has a new name, id, rates, and maximum amount that can be loaned for example. Administrators can also update existing loan policies and view all current loan policies within the system

2.5.2 Transaction Requirement: Data Entry

1. Enter new loan policies

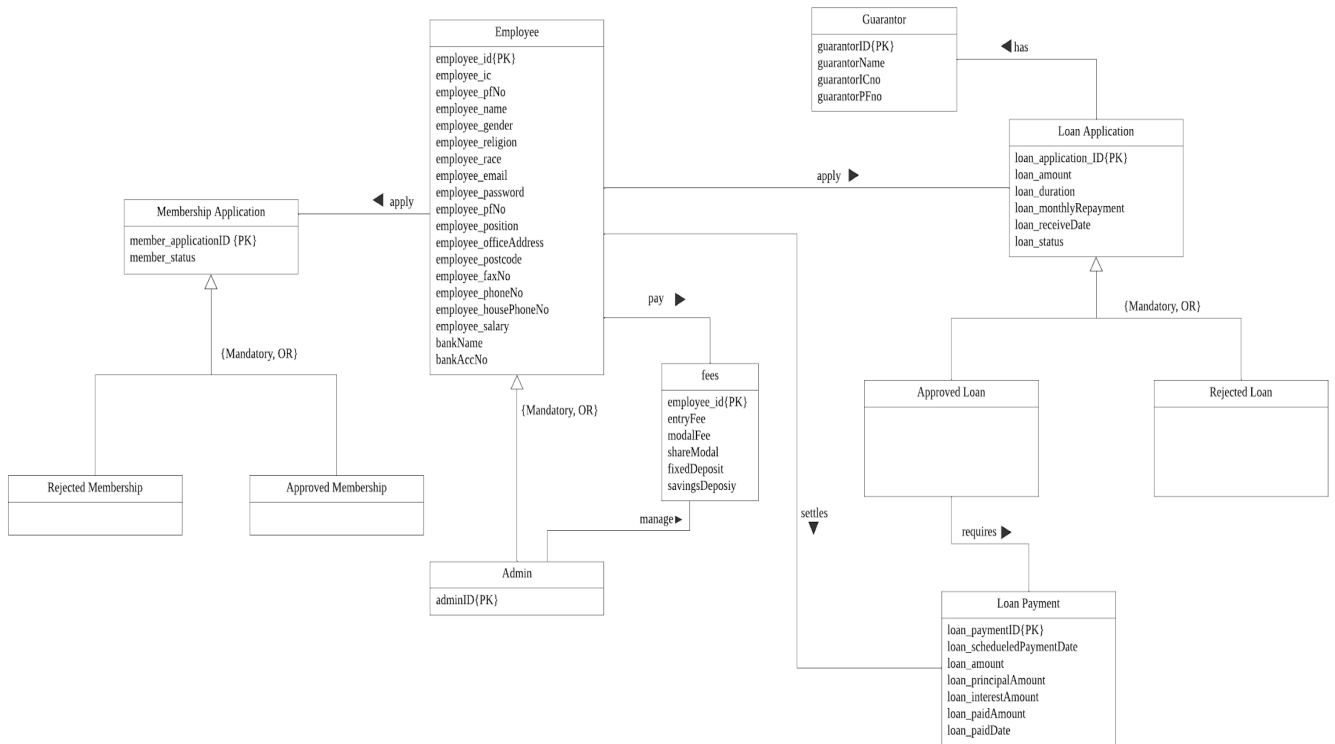
2.5.3 Transaction Requirement: Data Update/Deletion

1. Update loan policies
2. Update monthly fees of employees

2.5.4 Transaction Requirement: Data Queries

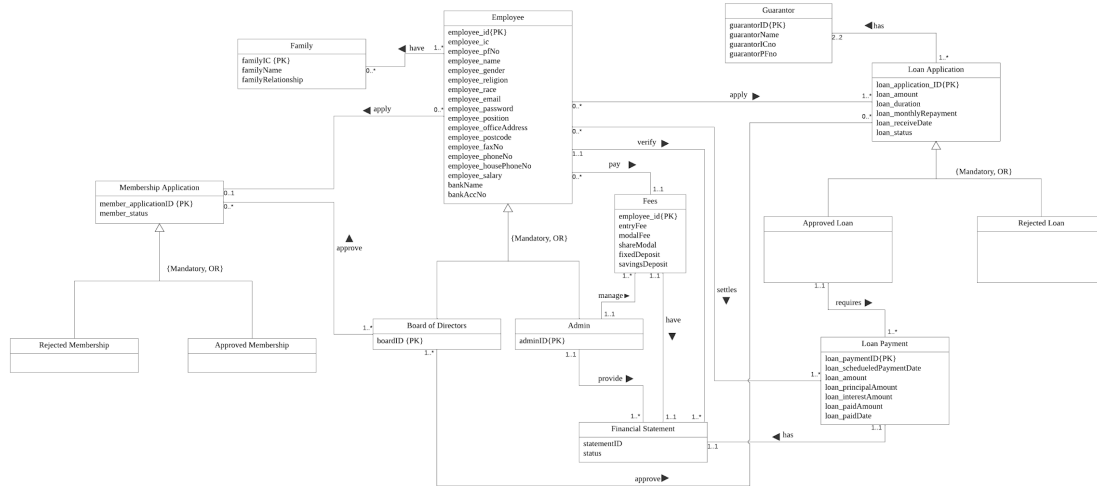
1. List of loan policies
2. List of monthly fees of each employee
3. List of membership application
4. List of employees and their details
5. List of loan applications

2.5.5 Local Conceptual Data Model (ERD) for Module 4

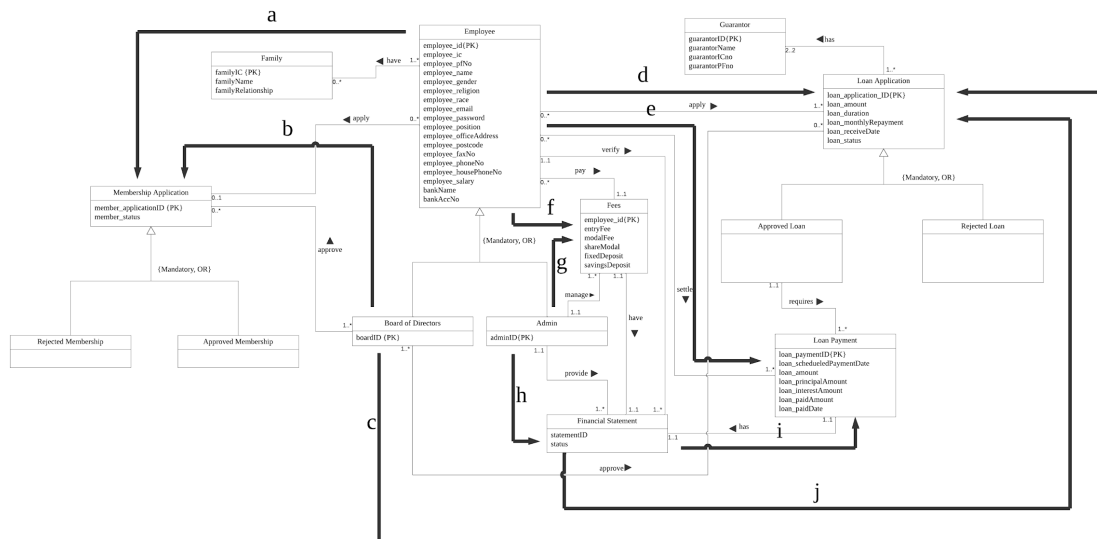


CHAPTER 3: Global Conceptual Data Model (Global ERD)

3.1 Global Conceptual Design (ERD)



3.2 Verified Global Conceptual Design (Verified ERD)



Symbol	Transaction
a	2.2.2.1, 2.2.3.1

b	2.2.3.2, 2.2.4.1, 2.2.4.2, 2.2.4.3, 2.2.4.4
c	2.3.3.1, 2.3.4.1, 2.3.4.2, 2.3.4.3, 2.3.4.4
d	2.3.2.1, 2.3.2.1, 2.3.3.2
e	2.4.2.2, 2.4.3.2
f	2.4.2.1, 2.4.2.3
g	2.4.3.1, 2.4.3.3
h	2.4.4.3, 2.4.4.5, 2.4.4.6, 2.4.4.7, 2.5.4.2
i	2.4.4.1, 2.4.4.2, 2.5.3.2
j	2.5.2.1, 2.5.3.1, 2.5.4.1

3.3 Data Dictionary for the global conceptual ERD

3.3.1 Entity Identification

Entity Name	Description	Aliases	Occurrence
Employee	General term describing KADA workers	Users	Occurs when employees apply for a loan and apply for membership.
Admin	General term describing staff of KADA Cooperative that manages cooperative databases and policies	Administrator, Supervisor	Occurs when admin want to manages cooperative databases and policies
Board of Directors	General term describing manager of KADA Cooperative	Manager	Occurs when board of directors need to approve loan and membership
Family	General term describing employee's family members	Employee household	Occurs when employee want to apply for loan
Guarantors	General term describing employee's guarantors	Warranters	Occurs when employee want to apply for loan
Membership Application	General term describing the process of employee want to apply for membership	Membership Request	Occurs when employee applying membership
Rejected Membership	General term describing employee failed to register as a member	Failed Membership	Occurs when employee failed to apply for membership
Approved Membership	General term describing employee success to register as a member	Successful Membership	Occurs when employee success to apply for membership
Fees	General term describing the amount of employees need to pay	Payment	Occurs when employee need to pay fee from

	for applying loan and membership		applying loan and membership
Financial Statement	General term describing the financial details that be sent to employee	Income statement	Occurs when admin need to change the financial status and employee when the want to view financial status
Loan Application	General term describing the process of employee applying for loan	Loan Request	Occurs when employee applying loan
Approved Loan	General term describing employee failed to apply for loan	Failed Loan Application	Occurs when employee failed to apply for loan
Rejected Loan	General term describing employee success to apply loan	Successful Loan Application	Occurs when employee success to apply for loan

3.3.2 Relationship Type Identification

Entity Name	Multiplicity	Relationship	Entity Name	Multiplicity
Employee	0..*	apply	Loan Application	1..*
	0..*	settles	Loan Payment	1..*
	1..1	verify	Financial Statement	1..*
	0..*	pay	Fees	1..1
	1..*	have	Family	0..*
	0..*	apply	Membership Application	0..1
Admin	1..1	manage	Fees	1..*
	1..1	provide	Financial Statement	1..*
Board of Directors	1..*	approve	Loan Application	0..*
	1..*	approve	Membership Application	0..*
Fees	1..1	have	Financial Statement	1..1
Loan Application	1..*	has	Guarantor	2..2
Approved Loan	1..1	requires	Loan Payment	1..*

3.3.3 Attribute Description

Entity Name	Attributes	Data Type & Length	Description	Nulls	Multi-valued	Example
Employee	employee_id{PK}	varchar (10)	Uniquely identifies each Employee	No	No	KADA0001
	employee_ic	int(12)	Employee Identity Card number	No	No	040429019078
	employee_pfNo	int(4)	Employee 'Personal File' number	No	No	1234
	employee_name	varchar (100)	Employee full name	No	No	AHMAD AKUB BIN LUKMAN
	employee_gender	char (M or F)	Employee gender	No	No	M
	employee_religion	varchar (10)	Employee religion	No	No	Islam
	employee_race	varchar (10)	Employee race	No	No	Melayu
	employee_email	varchar (100)	Email account of employee	No	No	ahmad12@gmail.com
	employee_password	varchar (100)	account password for login	No	No	mindacemerlang123
	employee_position	varchar (10)	Employee position in the company	No	No	Accountant
	employee_office Address	varchar (100)	Employee office address	No	No	1,Jalan Mawar,45670,Kelantan
	employee_postcode	int(5)	postcode of address	No	No	09123

	employee_faxNo	int(15)	Employee's fax number	No	No	6031234 5678
	employee_phoneNo	int(15)	Employee's phone number	No	No	01136718507
	employee_housePhoneNo	int(15)	Employee's house phone number	No	No	01987656781
	employee_salary	int(10)	Employee's latest salary	No	No	5000
	bankName	varchar (20)	The bank name of employee	No	No	Bank Kerjasama Rakyat
	bankAccNo	int	The bank account number of employee	No	No	126572821879811
Membership Application	member_applicationID {PK}	varchar (10)	Uniquely identifies employee that have registered for membership	No	No	KK10234
	member_status	varchar (10)	Status of membership	No	No	Active
Board of Directors	boardID {PK}	varchar (10)	Uniquely identifies Board of Director	No	No	BD001
Admin	adminID {PK}	varchar (10)	Uniquely identifies Admin	No	No	AD001
Fees	employee_id {PK}	int	Uniquely identifies each Employee	No	No	KADA0001
	entryFee	double	Entry fee that needs to be paid monthly by members	No	No	50.00
	modalFee	double	Modal that needs to be	No	No	50.00

			paid monthly by members			
	shareModal	double	Share modal that needs to be paid monthly by members	No	No	50.00
	fixedDeposit	double	Fixed deposit that needs to be paid monthly by members	No	No	50.00
	savingsDeposit	double	Savings deposit that needs to be paid monthly by members	No	No	50.00
Family	familyIC {PK}	int(12)	Identification number of an employee's family member	No	No	040429019078
	familyName	varchar (100)	Name of an employee's family member	No	No	AHMAD AKUB BIN LUKMAN
	familyRelations hip	varchar (10)	Employee's relationship to said family member	No	No	Father
Financial Statement	statementID	varchar (10)	An id that uniquely identifies statements	No	No	A123GRTRTY
	status	text (10)	Status to identify whether the financial statement is verified or not	No	No	Verified
Loan Application	loan_application _ID{PK}	varchar (10)	An id that uniquely identifies loan applications	No	No	A123GRTRTY
	loan_amount	double	Amount	No	No	10 000.00

			requested for a loan			
	loan_duration	int	Period of time the loan in scheduled for the loan payment in months	No	No	12
	loan_monthlyRepayment	double	amount of money required to paid monthly based on the loan amount and duration	No	No	300.00
	loan_receiveDate	date	date of loan application	No	No	20/12/2022
	loan_status	text (10)	the status of loan application whether it is rejected or approved	No	No	Approved
Loan Payment	loan_paymentID {PK}	varchar (10)	an id that uniquely identifies each payments made	No	No	A123GRTRTY
	loan_scheduledPaymentDate	date	date of each month where the payments should be made by then	No	No	20/12/2022
	loan_amount	double	Amount requested for a loan	No	No	10 000.00
	loan_principalAmount	double	the current loan amount	No	No	9 000.00
	loan_interestAmount	double	the total interest from the loan amount	No	No	1 500.00

	loan_paidAmount	double	the paid amount of loan	No	No	1 000.00
	loan_paidDate	date	the date where the payment is made	No	No	3/1/2024
Guarantor	guarantorID{PK}	varchar (10)	an id to uniquely identify guarantors	No	No	A123GRTRTY
	guarantorName	varchar (100)	the name of guarantor	No	No	AHMAD AKUB BIN LUKMAN
	guarantorICno	int(12)	Guarantor identity card number	No	No	040429019078
	guarantorPFno	int(4)	Guarantor 'Personal File' number	No	No	1234

CHAPTER 4: Appendices

4.1 Use Case Diagram for the system.

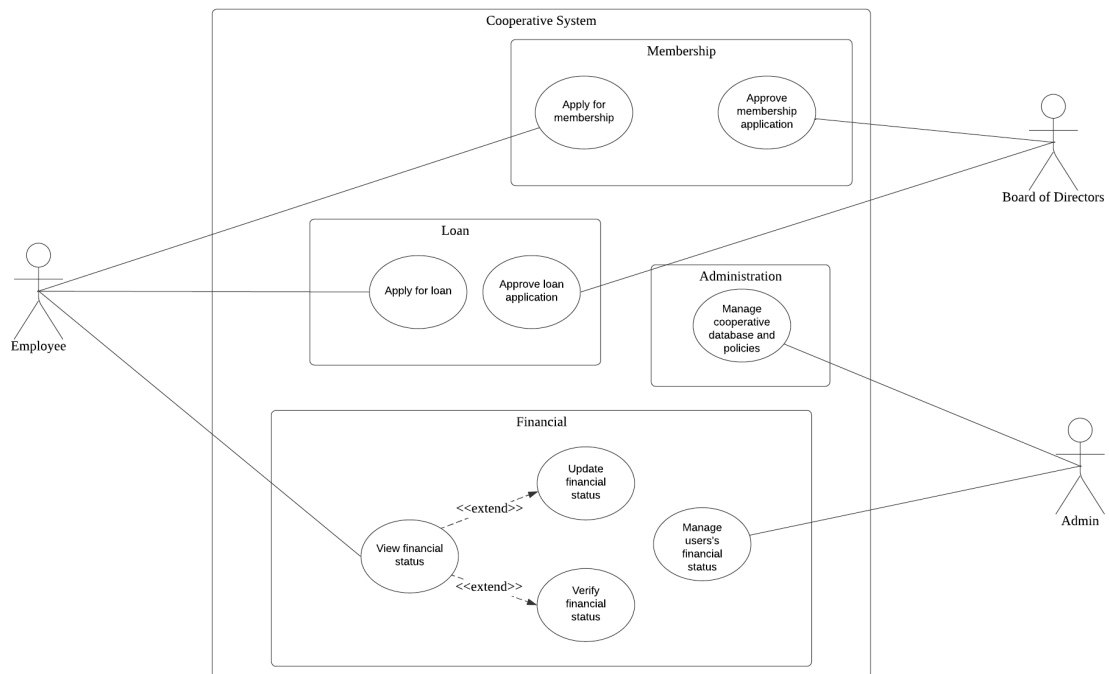
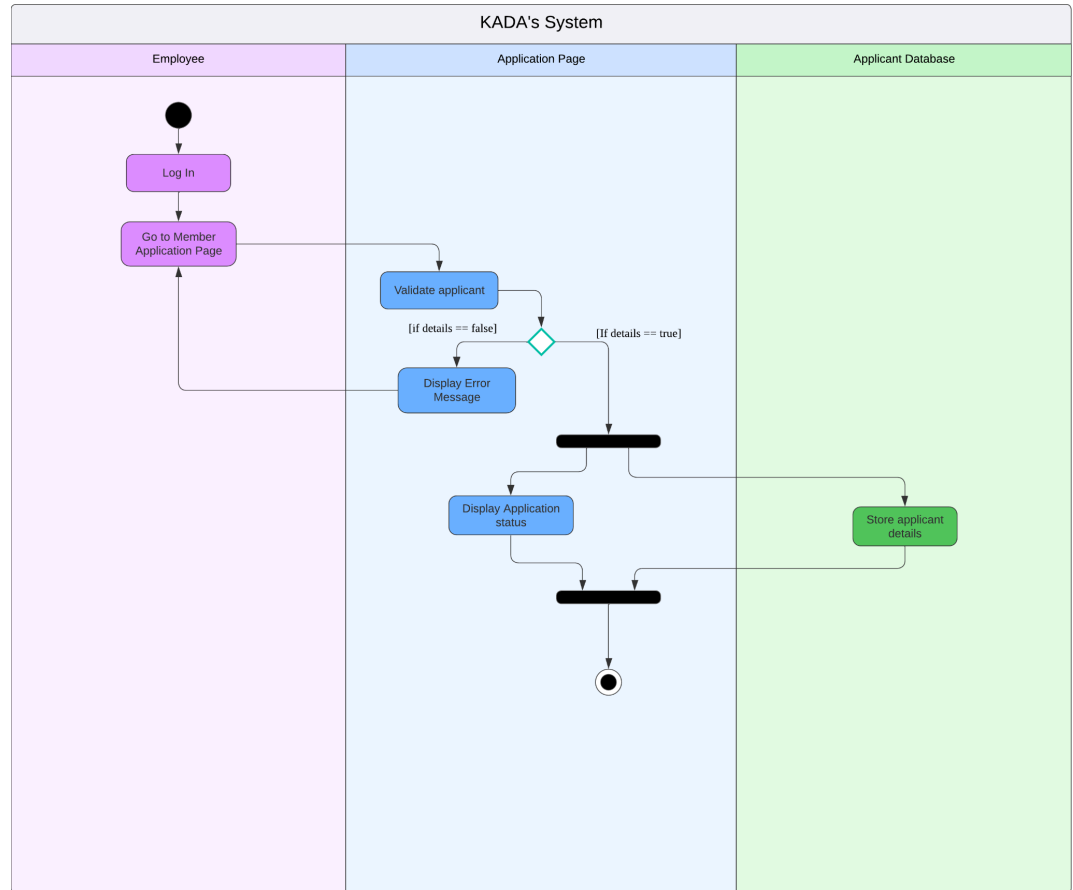


Figure 4.1 Use Case Diagram of the System

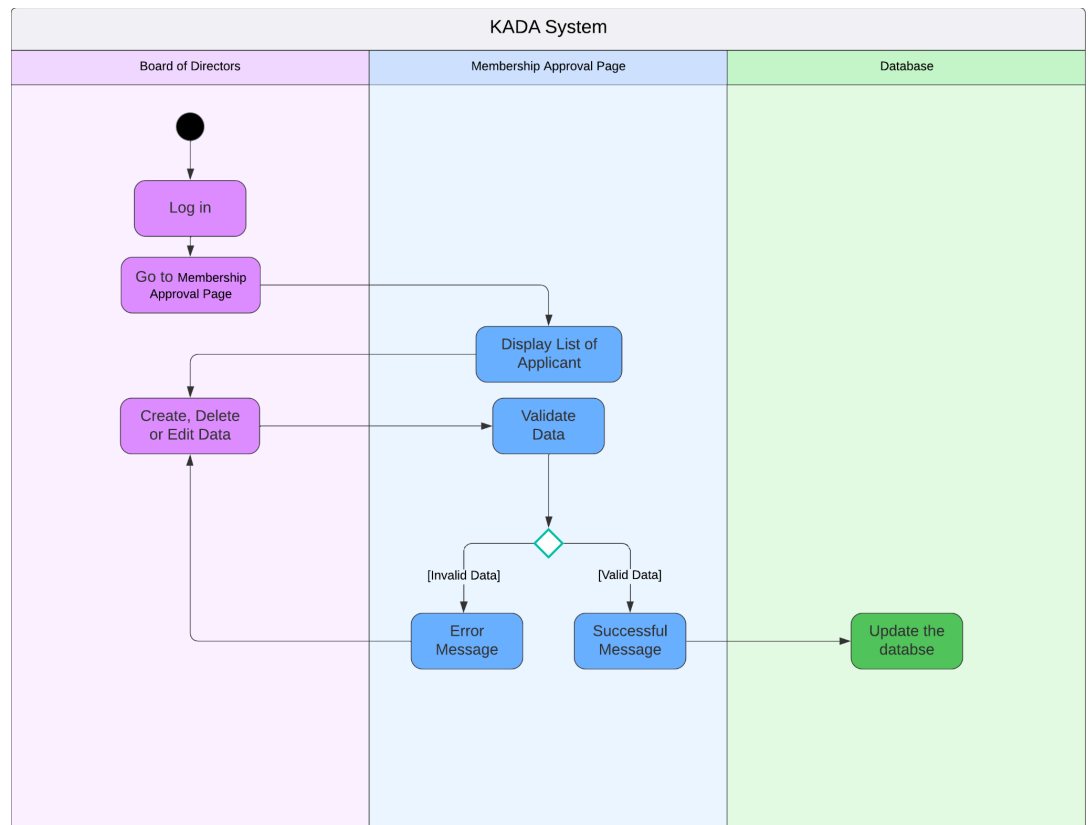
4.2 Activity Diagrams for every use case identified.

4.2.1 Module <Membership>

4.2.1.1 Apply for Membership

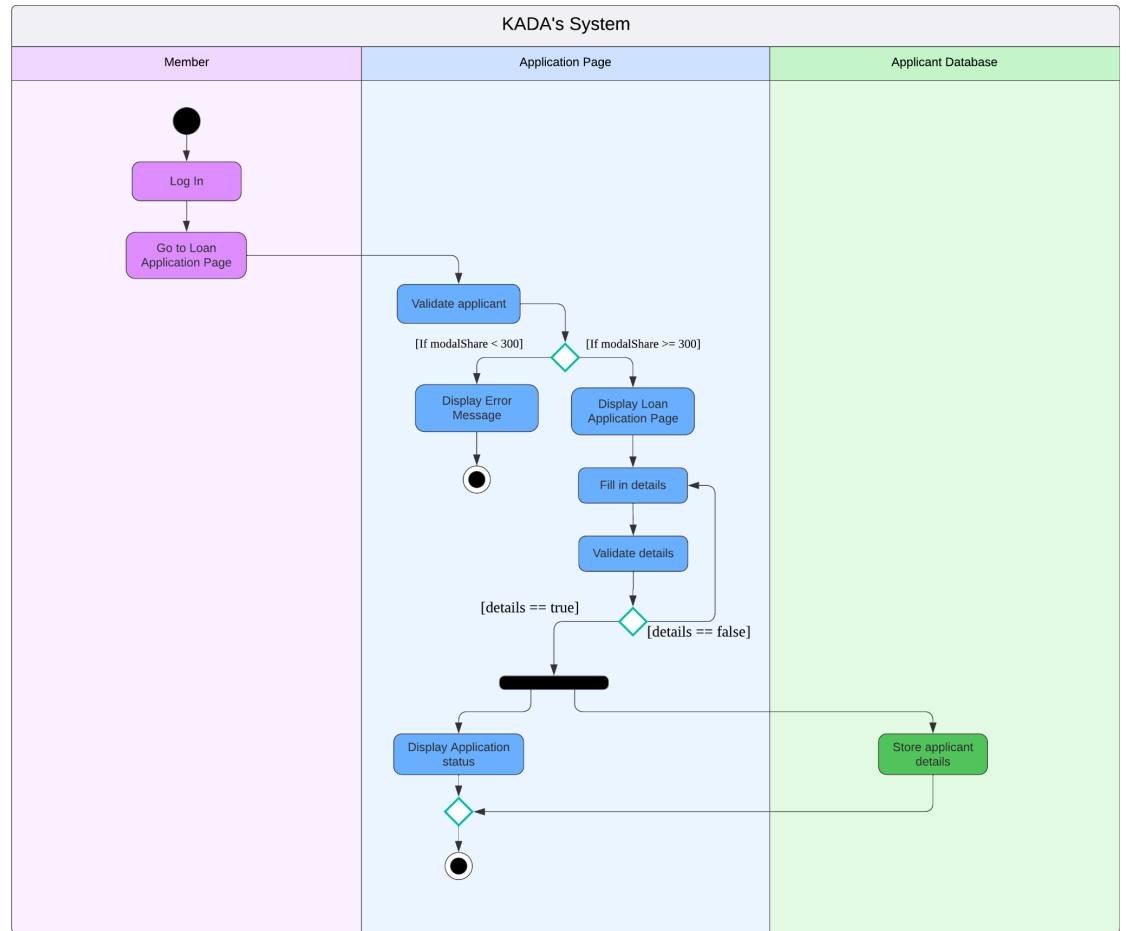


4.2.1.2 Approve Membership Application

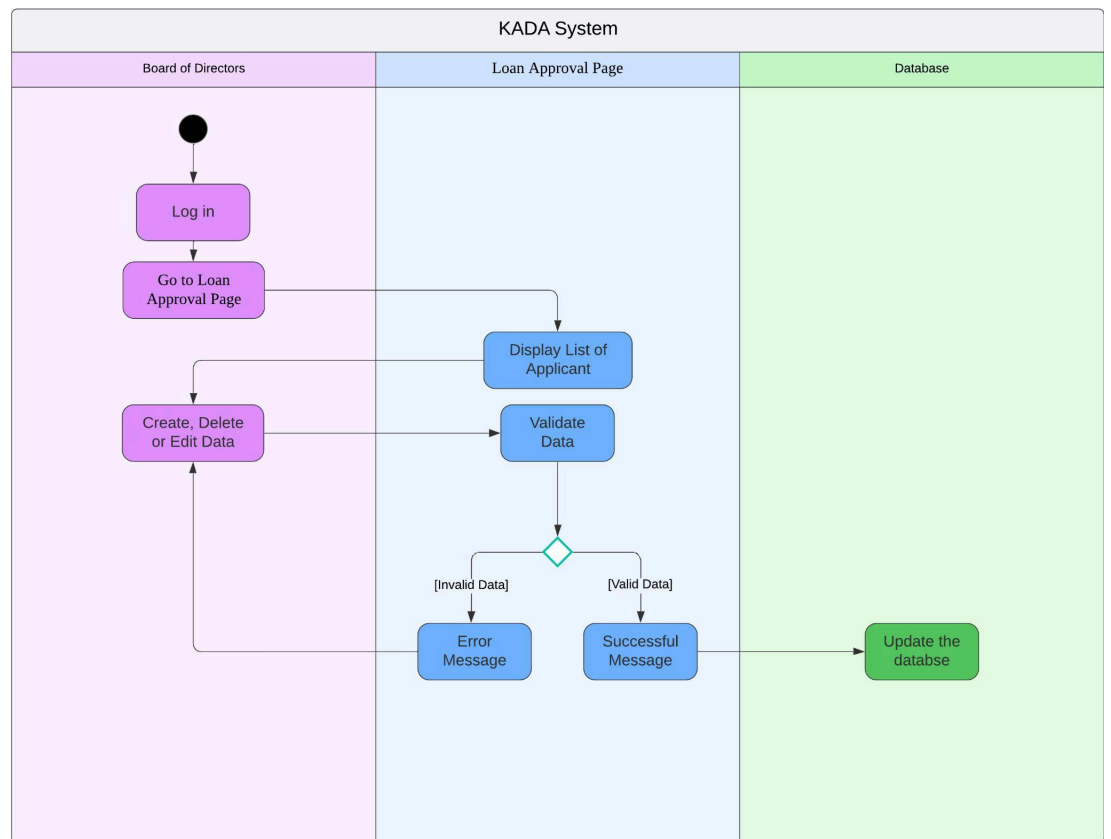


4.2.2 Module <Loan>

4.2.2.1 Apply for Loan

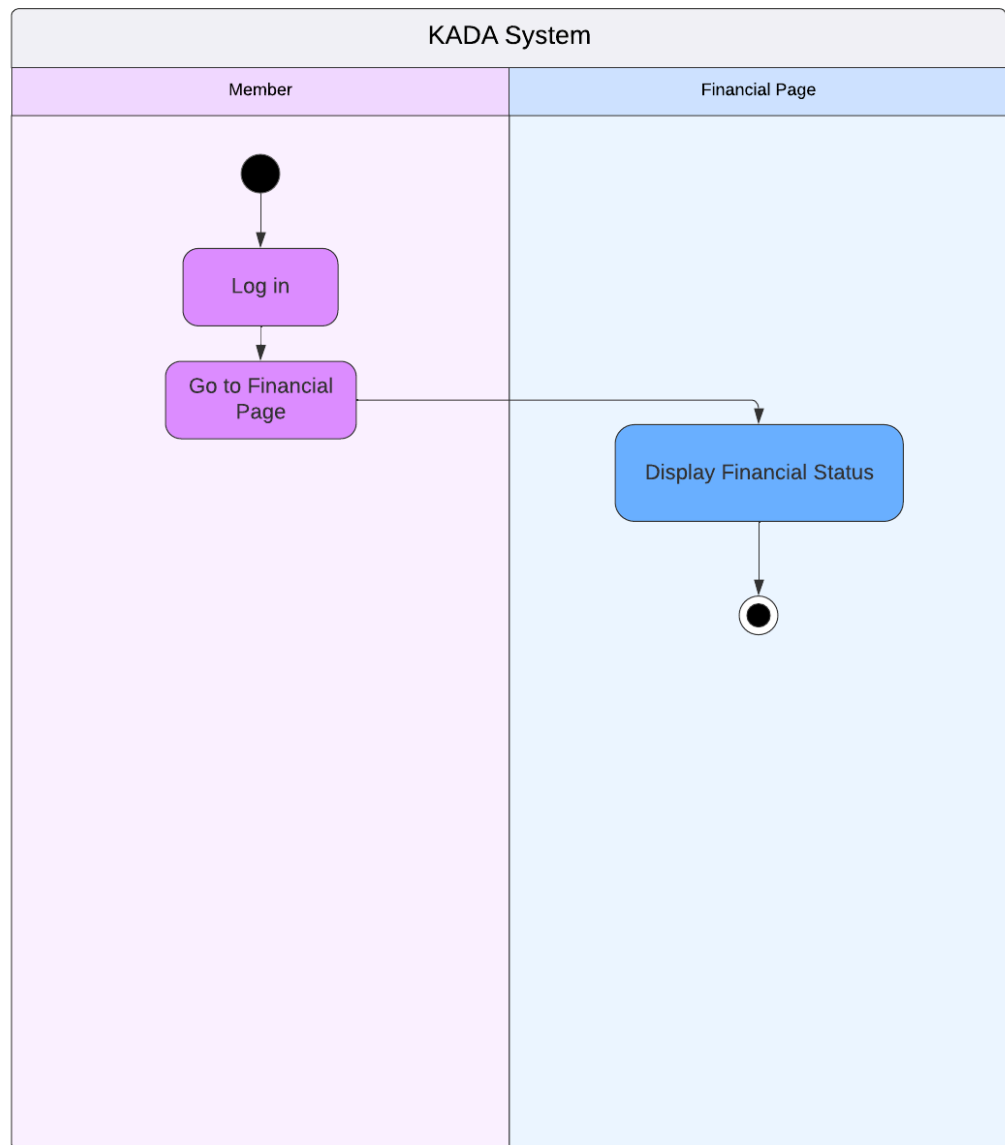


4.2.2.2 Approve Loan Application

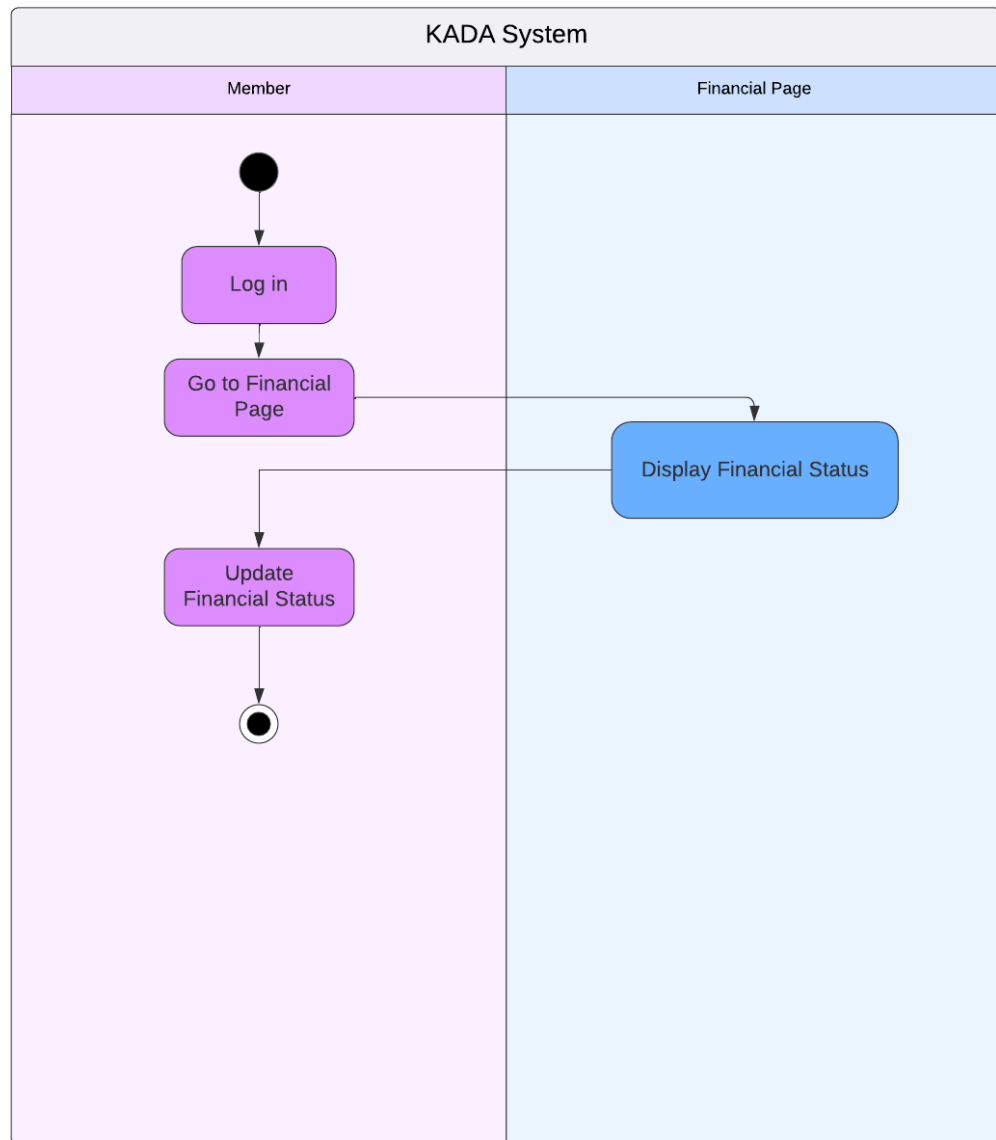


4.2.3 Module <Financial>

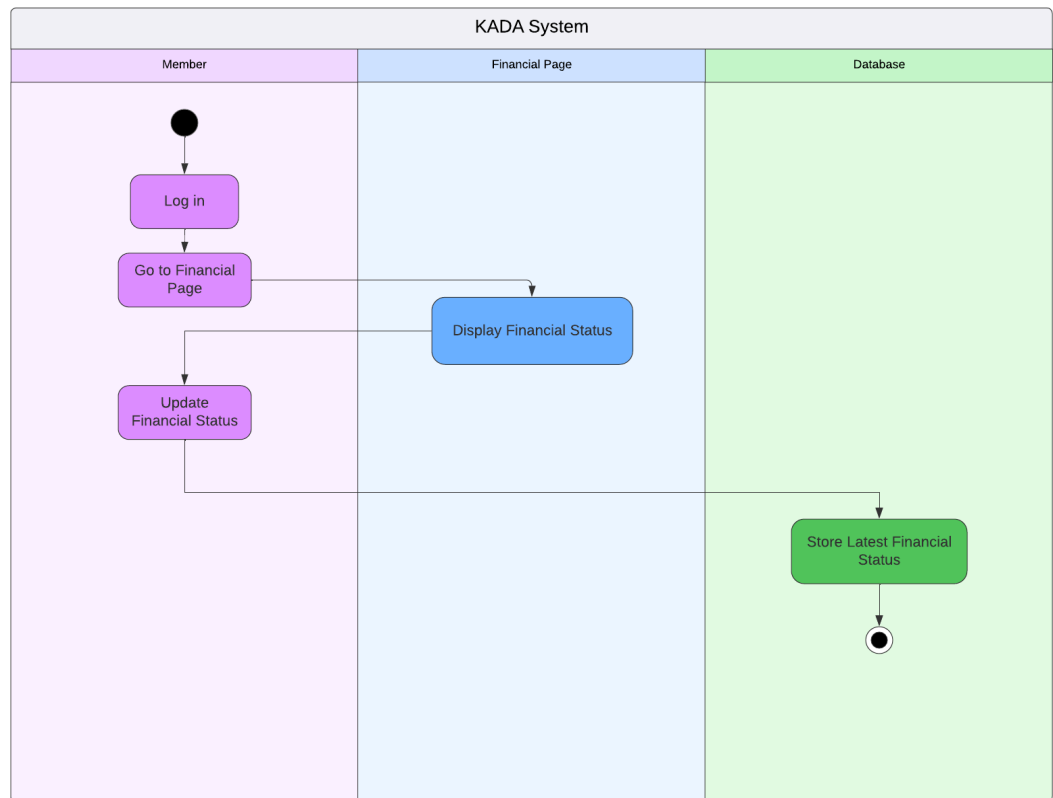
4.2.3.1 View Financial Status



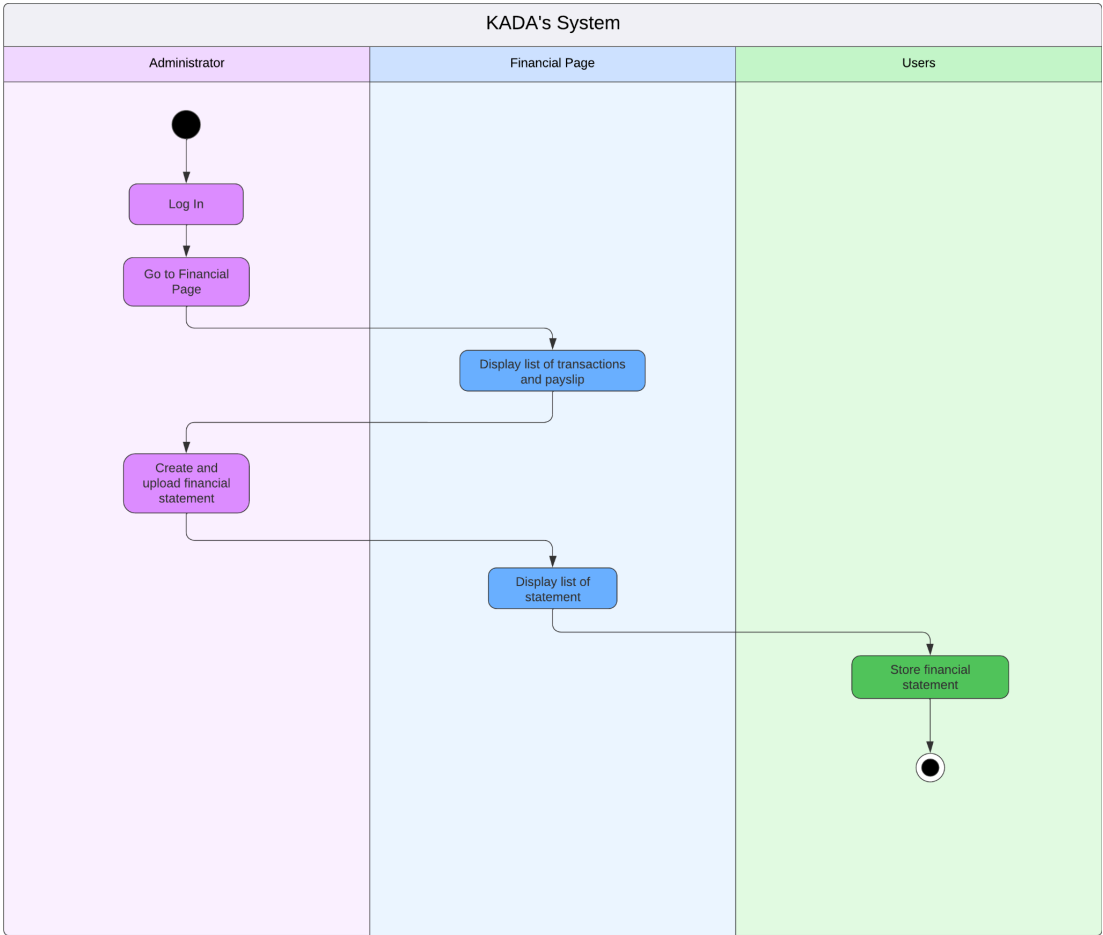
4.2.3.2 Update Financial Status



4.2.3.3 Verify Financial Status

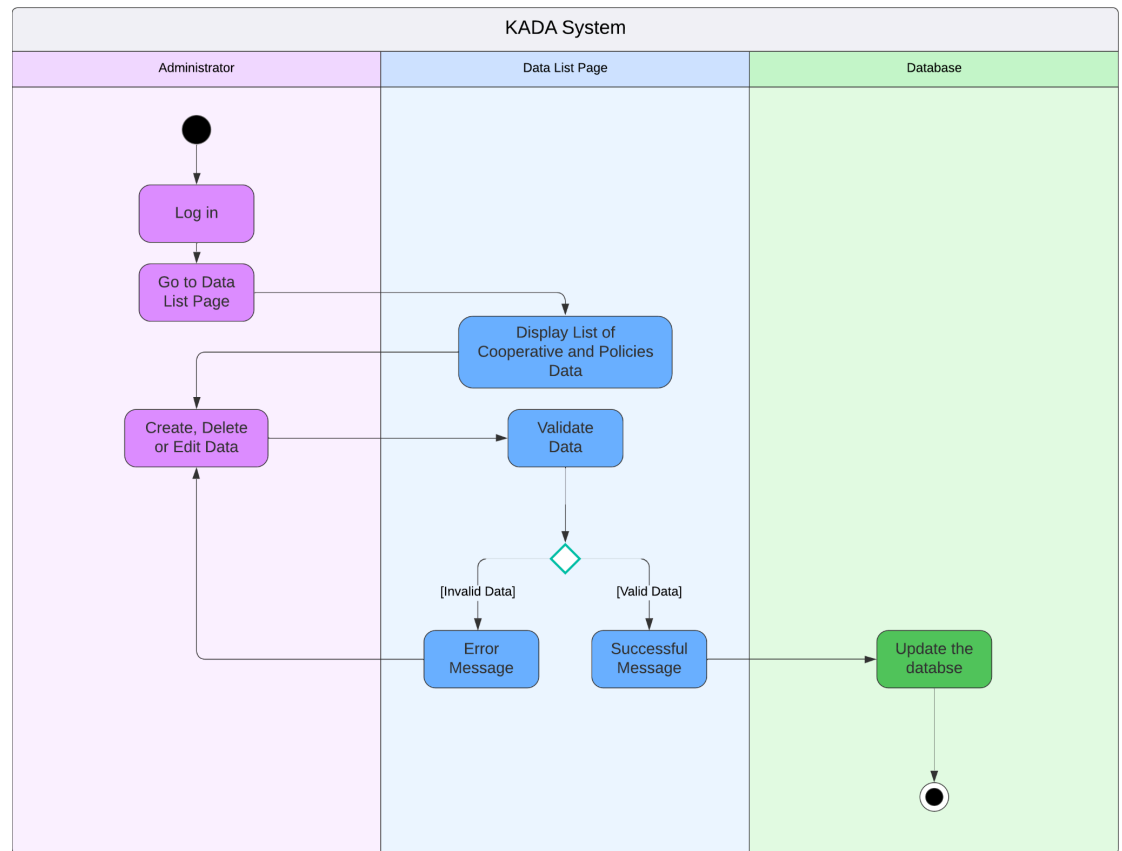


4.2.4.4 Manage User's Financial Status



4.2.4 Module <Administration>

4.2.4.1 Manage Cooperative Databases and Policies



4.3 Meeting Log

We initiated our meeting on 12 December 2024, Thursday around 9.00 pm. We start with the discussion among the group about the specific task that has been assigned to us. Fortunately, all members participated during the meeting and none of us were absent. Our priority is to contribute some ideas to other member and also divide tasks to a small chunk for each team member equally.

We ended our discussion around 10.00 pm on 16 December 2024, Tuesday. We have concluded everything from the start until the end. Each team member dedicated and continue to share their own contributions all throughout the project phase 2.